

SEQUENTIAL CANDIDATE SELECTION FOR SCORE FOLLOWING IN SHEET MUSIC IMAGES

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ABSTRACT

Score following aligns a live performance to its score in real time, committing to a position at every instant with no lookahead. Its two sides are different kinds of object, the performance audio and the score an image, and the established response is to convert one into the other and align within a single modality, by optical music recognition or by transcription, each detour inheriting the errors of its conversion. Sheet-image score following converts neither, conditioning a visual detector on the audio, and what limits it is annotation, since alignments to notehead coordinates are costly enough that nearly all training data is synthesized and these systems lose much of their accuracy on a real instrument in a real room. Every remedy proposed for that gap has treated it as a failure of perception. We show that it is not, since the detector already proposes a position within the standard tolerance at every scored onset of a real recording and an oracle held to the same realtime commitments recovers almost everything the deployed rule loses. We therefore recast the task as sequential selection over the candidates the detector already emits, combining a parameter-free transition model with a set-based scorer that judges each candidate against its competitors. This sets a new best on real piano recordings and shows that what the decision needs is its own temporal history rather than a better image representation.

Index Terms— Score following, sheet music, audio-to-score alignment, realtime, music information retrieval

1. INTRODUCTION

Music alignment puts a performance and its score in correspondence, mapping each moment of the audio to a position in the written music. The two regimes differ in whether the alignment may look ahead. Offline, where the task is known as audio-to-score alignment or music synchronisation, the whole recording is in hand, a global method such as dynamic time warping settles every position at once [1], repeats and jumps are recoverable after the fact [2, 3], and the correspondences support cross-modal retrieval and piece identification [4].

Online, none of that is available. The system observes only what has been played, commits immediately and never revises. This is score following [5], also called music tracking [6], in which the audio up to frame t determines the position the performer currently occupies and the estimate must be produced before frame $t + 1$ arrives. Automatic page turning [7], live computer accompaniment [8, 9] and score-synchronised display all depend on it, and every result below, the oracles included, respects the constraint.

What makes it hard is that the two sides are not the same kind of object. The performance arrives as audio and the score exists as an image, with no space in which to compare them directly, and the literature resolves this in three ways, distinguished by which side, if either, is converted.

1.1. Related work

The first converts the score. Optical music recognition renders the page as symbols, after which a classical follower tracks a discrete sequence of score states, a line running from online dynamic time warping [1] through duration models [10], hidden semi-Markov models [11], particle filters [12], switching state-space models [13] and multi-agent tracking [6]; recognition remains error-prone on real scores [14] and the follower inherits every mistake. The second converts the audio, and recent work pairs realtime piano transcription with symbol-level tracking to strong effect [15], at the price of bounding the follower by the transcriber.

The third converts neither, learning the correspondence between audio and pixels directly. Dorfer et al. [16] classified position into staff-level buckets on crops; reinforcement learning framed tracking as an agent on a score strip [17, 18]; CUNet [19] emitted an audio-conditioned segmentation of the full page; and CYOLO [20] recast it as audio-conditioned object detection, CYOLO-SB [21] adding heads for the enclosing system and bar. CODA [22] cascades system, bar and note selection over layout regions with beam search and learned transition penalties and is the strongest published system here, though it reads the page’s system and bar boxes at test time, which neither the others nor ours require.

All three depend on performances aligned to notehead coordinates, an annotation costly enough that nearly all training data is synthesized, so systems learn on rendered audio and

lose much of their accuracy on a real instrument in a real room, CUNet falling from 85.5% of onsets within 0.5 s to 22.4%, CYOLO-SB from 89.3% to 79.9% and CODA from 95.5% to 88.3%. Every remedy offered has been a perception remedy, whether reverberant augmentation [20], extra proprietary recordings [21] or a new architecture [22].

1.2. Diagnosis

We tested that premise before adopting it. An audio-conditioned detector does not emit a position; it scores thousands of candidate noteheads and reports the most confident of them. Retaining the 256 highest-scoring candidates, we find that on real room-microphone recordings one of them lies within the 0.5 s tolerance at every one of 4,149 scored onsets, so an oracle free to choose among them is exact, though no such oracle can run live. The one that can must commit at each instant and never move backwards, as a deployed follower must, and it still reaches 99.6% where the deployed rule reaches 80.0% (Fig. 1). Nineteen of the twenty points lost to real audio survive the realtime constraint. At this tolerance the room degrades the detector’s ordering of its own candidates more than the visual evidence underneath, and Sec. 4 shows how far that holds as the tolerance tightens.

That reading argues for treating the task as a sequential decision over the positions a page affords rather than as the regression of a coordinate. CANDOR makes that decision from a transition model, which scores a displacement against the time elapsed, and a set-based scorer, which judges each candidate against its competitors in the same instant. It is trained listwise on synthesized audio alone and decodes causally and greedily, at 1.25 ms per instant on one CPU thread against the 50 ms available at 20 frames per second.

Our contribution is to relocate the synthetic-to-real gap and to act on it. Candidate-set oracles place the gap in the decision rather than in perception, and a decision model built on that finding sets a new best on real recordings, by a margin that holds across three released detectors differing in heads and in training data.

2. METHOD

2.1. Backbone

Every sheet-image follower of the last five years shares one perception module, a convolutional network over the score page whose feature maps are modulated by the audio. The mechanism is feature-wise linear modulation [23], which injects an external vector into a network by scaling and shifting its activations, $f_{\text{FiLM}}(x) = s(z) \odot x + t(z)$, where s and t are learned linear maps of a conditioning vector z and $\odot$ is elementwise. The page alone cannot say where the performer is, so z carries the recent audio and the visual features become a function of what has just been heard.



Fig. 1. One instant of a real room-microphone recording. In yellow the 64 most confident candidates, darker where several share a notehead. The most confident (red) is on the wrong staff, 2.3 s off; CANDOR (teal) takes a less confident one on the true note (ring), forward of the previous decision (caret). The correct position was available and was not chosen.

CYOLO [20] encodes the last two seconds of audio into z with a spectrogram network and places FiLM layers through the downscale and upscale blocks of a YOLO detector [24], which then predicts a bounding box for the notehead currently sounding. CYOLO-SB [21] keeps that arrangement and adds heads for the enclosing system and bar, so one 52×52 feature map yields all three. CODA [22] conditions its own backbone the same way, at the deeper stages, before cascading selection over layout regions supplied with the score.

We take this module as given rather than as a design choice, and use the released CYOLO detectors to propose noteheads. No detector weight is updated anywhere in this work, and three released instances are used, differing in their heads and in their training data, so the conclusions concern the family and not one checkpoint.

2.2. CANDOR

CANDOR, for candidate ordering, sits after the backbone and replaces only its final step. At each instant the detector offers a few hundred possible notehead positions, and where it would report the most confident of them, CANDOR asks which one is most compatible with where its own previous decision was and with what the rest of the set looks like. Those two questions are the two parts below, and answering them requires carrying state from one instant to the next.

2.2.1. Candidate positions

A performance is processed in instants $t = 1, \dots, T$, and at each one the backbone reads the audio so far and the current page and scores note boxes $b_i = (x_i, y_i, w_i, h_i)$ with confidence o_i ; we keep the $K = 256$ most confident, C_t . Staves are concatenated along one horizontal axis, the unrolled coordinate u_i , so positions across a line break are adjacent. The released detectors score 8,112 boxes per instant and report the most confident, $\hat{u}_t = \arg \max_i o_i$, which is the degenerate selector. CANDOR reads C_t and its own previous choice $\hat{u}_{t-1}$,

and nothing else. K is set where availability stops binding, since a correct candidate is present at every onset only from 256 and the causal oracle falls to 85.6% at $K = 64$, below what CANDOR itself reaches, so a smaller set would limit the decision rather than test it.

2.2.2. Transition model

Selections must form a plausible trajectory, and between instants Δ_t frames apart the performer advances by a roughly predictable amount, so the displacement $d_i = u_i - \hat{u}_{t-1}$ is scored by a floored Gaussian whose mean grows with elapsed time,

$$\log \pi_i = \max \left(-\frac{(d_i - \mu \rho_t)^2}{2\sigma^2}, J \right), \quad (1)$$

with $\rho_t = \text{clip}(\Delta_t/\Delta_0, 0.2, 8)$ and $\Delta_0 = 5$ frames, the mean inter-onset gap in training. The floor J keeps a jump possible at bounded cost. Combined with confidence, $h_i = \log o_i + \log \pi_i$ already recovers a large part of the gap on its own.

2.2.3. Set-based scorer

The transition model treats candidates independently given the history, so what it cannot express is comparative, whether this box, among those on offer in this instant, looks like a real notehead rather than a spurious response. Real audio calls for exactly that, since reverberation smears the onset the conditioning vector is built from and the detector answers with many moderately confident boxes rather than one sharp response, leaving absolute confidence poorly calibrated while the shape of the set stays informative. Each candidate is described by $\phi_i \in \mathbb{R}^{33}$ holding its confidence relative to the instant's best, box geometry, motion relative to the history, relation to any coarse regions the backbone emits, and how crowded and how typical the box is among its neighbours, together with f_i , the 128-dimensional FiLM-conditioned feature the detection head scored the box from. A permutation-invariant network [25], as in learned re-scoring of detections [26], embeds each candidate as $e_i = g([\phi_i; \text{ReLU}(Wf_i)])$ and scores it against a context pooled over the set, $s_i = r([e_i; c])$ with $c = [\frac{1}{K} \sum_j e_j; \max_j e_j]$, the maximum taken per dimension. Here g and r are two-layer perceptrons (64, 32 and 64, 1) and $W \in \mathbb{R}^{64 \times 128}$, so every candidate is judged against its competitors.

2.2.4. Training and inference

The decision is supervised listwise against soft targets, so with ε_i the error in frames of candidate i the target is $q_i \propto \exp(-\varepsilon_i/\tau)$ with $\tau = 3$ and the loss is the cross-entropy $-\sum_i q_i \log \text{softmax}(s)_i$, so the scorer learns to order a set rather than to regress a coordinate. History features come from previous decisions, perturbed on 30% of samples by Laplace

noise of scale 30 px to expose training to the drift of sequence prediction [27]. At inference CANDOR decides

$$\hat{i}_t = \arg \max_{i \in \mathcal{C}_t} \beta s_i + (1 - \beta) h_i, \quad (2)$$

causally and greedily, one pass over 256 candidates per instant, in 1.25 ms on a single CPU thread against the 50 ms available at 20 fps. Training on the residual $s_i - h_i$ and deciding $\arg \max_i (s_i + h_i)$ is Eq. 2 at $\beta = 0.5$ and eliminates all four constants.

3. EXPERIMENTAL SETUP

3.1. Dataset

All experiments use MSMD [4], which pairs public-domain LilyPond engravings with MIDI-derived alignments for solo piano. Its repertoire is Western classical, weighted towards the Baroque and Classical and dominated by Bach, with Beethoven, Mozart, Handel, Schumann, Chopin, Czerny and Burgmüller making up most of the remainder, and textures from two-voice inventions to dense Romantic writing, which sets how many plausible candidates a page affords at each instant.

The decision model is trained on 353 pieces, synthesized and convolved with measured room impulse responses [28], and validated on 19, using AdamW ($3 \cdot 10^{-3}$, cosine, 40 epochs, batch 256). Evaluation is the real-performance benchmark of [18], 16 pieces of Bach, Schumann, Mozart, Chopin and Mussorgsky spanning 25 pages and 4,149 onsets, played on a Yamaha AvantGrand N2 and captured with a room microphone, with the correct page given throughout. A further 80 held-out pieces are rendered with a real impulse response and noise from 12 to 0.5 dB SNR. We report the percentage of onsets whose predicted position maps to within 0.05 to 5 s of the true time, 0.5 s primary, with 95% intervals bootstrapped over pieces [29] and seed means with deviations.

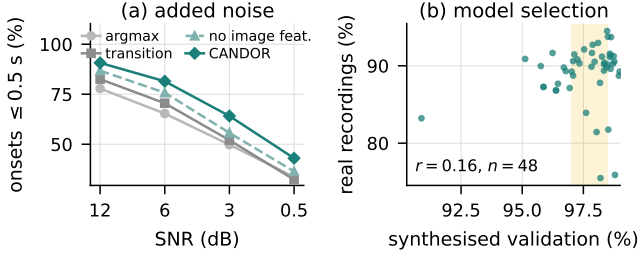
Three kinds of data enter at three stages. The decision model is trained on synthesized audio alone and never sees a real recording, the four constants (μ, σ, J, β) are calibrated leave-one-piece-out on the other 15 real pieces weighted equally, and evaluation is on the piece held out. That still lets the benchmark as a whole participate in choosing them, which the published baselines did not require, so the residual form of Eq. 2 is carried alongside as the variant fitting nothing to any real recording.

4. RESULTS

CANDOR orders proposals, so it applies to a follower that produces a candidate set and then discards it. CODA [22] is not such a target, since its final stage regresses a continuous bar-local coordinate rather than choosing among proposals, and the sets it does select over, systems and bars, are supplied

Table 1. Real room-microphone recordings. [†] proprietary training data, [‡] layout annotation at test time.

Method	Onsets tracked (%) within threshold (s)					Δ
	≤ 0.05	≤ 0.1	≤ 0.5	≤ 1.0	≤ 5.0	
MM-Loc [16]	36.4	40.6	58.5	61.1	73.5	
RL agent [18]	18.5	20.0	47.6	60.3	90.1	
CUNet [19]	11.3	12.5	22.4	26.6	44.3	
CYOLO [20]	56.3	58.1	71.2	74.9	91.9	
CYOLO-SB [21]	61.0	63.0	79.9	83.2	96.0	
CYOLO-SB+A [†] [21]	68.2	70.6	86.5	89.1	98.1	
CODA [‡] [22]	72.1	74.3	88.3	91.4	98.7	
CANDOR _{CYOLO}	72.6	74.7	91.6	95.0	99.1	+20.4
CANDOR _{CYOLO-SB}	69.0	71.4	90.6	94.2	98.2	+10.7
CANDOR _{CYOLO-SB+A[†]}	76.0	78.5	94.4	96.9	99.6	+7.9
<i>Detectors we retrained, 24 h, audio tower only</i>						
CANDOR _{CYOLO-LSTM} (control)	58.2	60.6	83.3	87.5	98.3	+11.6
CANDOR _{CYOLO-Mamba}	62.8	65.7	91.5	94.6	99.3	+19.0
CANDOR _{CYOLO} , audio tower as [22]	44.7	47.4	76.2	83.9	96.5	+17.3

**Fig. 2.** (a) 80 held-out pieces under a room impulse response and additive noise, 40,594 onsets per level. (b) 48 trained decision models; half fall in the shaded 1.5-point validation band and differ by 19 points on real audio.

by layout annotation and already decoded with beam search. It appears in Table 1 as the strongest published system on these recordings, not as a backbone CANDOR could be attached to.

Table 1 sets CANDOR against the published literature, taken from [22], on recordings played straight through, with Δ the gain at 0.5 s over the perception model each row is built on. CANDOR tracks 94.4% of onsets within 0.5 s, 95% CI over pieces [92.2, 97.1], against a previous best of 88.3%, and leads at every threshold. Sixteen pieces are few and the interval is correspondingly wide, though the gain over the argmax it replaces, paired on the same resample, is +7.9 with interval [+2.8, +12.2]. Nor is the margin an artifact of one perception model. Retrained on whichever candidates each emits, the decision gains 20.4, 10.7 and 7.9 points, and on the weakest, whose training data is entirely public, it passes where the strongest one begins. A decision model does not need good proposals, only proposals containing the answer. On 94 synthetic test pieces it reaches 95.5% against 89.3% for argmax, and on the 80 noisy held-out pieces it beats the transition model at every level from 12 to 0.5 dB (Fig. 2a).

Table 2. Components of CANDOR on the real recordings at 0.5 s.

	n	argmax	+transition	full
<i>Perception model</i>				
CYOLO [20]	3	71.3	72.5	91.6 $\pm$ 1.4
CYOLO-SB [21]	6	80.0	86.5	90.6 $\pm$ 0.9
CYOLO-SB+A [†]	3	86.5	90.8	94.4 $\pm$ 0.9
<i>Decision model, CYOLO-SB candidates</i>				
residual, no fitted constants	3			88.7 $\pm$ 0.9
without temporal history	3			83.7 $\pm$ 0.4
<i>Image features f_i, CYOLO-SB candidates</i>				
FiLM-conditioned backbone	6			90.6 $\pm$ 0.9
DINOv2 [30]	1			90.9
none	1			90.9

The tightest thresholds qualify this. On CYOLO-SB candidates CANDOR reaches 69.0% and 71.4% at 0.05 and 0.1 s, below CODA’s 72.1% and 74.3%, since a decision is only as precise as the boxes it chooses between and the mechanism recovers which notehead is active rather than the position within it.

Table 2 separates the parts over the same candidates, adding Eq. 1 to the argmax and then the scorer, with n counting decision-model seeds at a fixed detector. The transition model alone, with no learned parameters, accounts for roughly half the gain. The residual variant is the one to read for the strictest protocol, fitting no constant to any real recording and so comparable with the published rows on equal terms; it reaches 88.7% against the 80.0% it replaces, above the previous best, and the distance from there to 94.4% is what calibrating four constants is worth.

4.1. Ablation study

Giving the scorer the FiLM-conditioned backbone feature, DINOv2 patch features [30] on the same grid, or no image feature at all yields 90.6, 90.9 and 90.9, a spread well inside the 0.9-point seed deviation, where on synthesized validation audio the same three differ by 3.4 points. Removing the features computed against the previous decision instead costs 7.2 points, the largest single ablation here, and teacher forcing attributes two thirds of the remaining error to drift, since given the true previous position the same scorer reaches 98.0% against the causal oracle’s 99.6%.

Retraining the detector itself from scratch on a fixed 24-hour budget with a different audio encoder gives the lower block of Table 1, not comparable with the converged checkpoints above it and reported because the decision gains a similar margin whatever it is attached to. A frozen MERT-95M tower [31] reaches 46.2 and overfits 12.6-fold; the two-layer causal Mamba [32] of [22] in place of the whole audio tower reaches 76.2 against 83.3 for an identically budgeted control; and keeping the convolutional front end while replacing only the recurrence, which we call CYOLO-Mamba, reaches 91.5. The recurrence therefore benefits from a state-space model while the convolutional front end is load-bearing, which is the encoder ablation absent from [22]. Replacing the visual stem with frozen DINOv2-base patches on the same 52×52 grid, against an identically retrained control, gives .. None of this rescues the gap, the strongest of these detectors merely reaching what re-deciding over an already released one achieves.

Across 48 trained models, accuracy on the synthesized validation split correlates $r = 0.16$ with accuracy on the real recordings (Fig. 2b), and the best-validation model scores 89.3, below one reading no image features at all. A reverberant held-out set correlates more strongly ($r = 0.80$) yet still selects poorly. Since the field selects models on synthesized audio, this is a caution beyond our own experiments.

5. CONCLUSION

The synthetic-to-real gap in sheet-image score following has been read as a failure of perception and answered with better audio, more data and new architectures. The measurement reported here places it elsewhere. On real room-microphone recordings the detector already proposes a correct position at every scored onset, and an oracle held to the same realtime commitments recovers almost all of what the deployed rule loses, so at the standard tolerance what fails is the rule that reads a position out of the evidence rather than the evidence itself.

Acting on that, CANDOR orders the candidates the detector already emits and takes real piano recordings from 80.0% to 94.4% of onsets within 0.5 s, against 88.3% for the best published system, at 1.25 ms per instant on one CPU thread, without retraining any detector and without layout annotation,

by a margin holding across three released detectors. Within this framework the decision depends far more on its own temporal history than on the image representations we tested. Two results temper it. Accuracy on synthesized audio barely predicts accuracy on real audio across 48 trained models, which makes selection on a synthetic proxy unreliable for the field and not only for us, and the forward-motion assumption of Eq. 1 confines the method to performances played in the written order. Pitch agreement between a candidate and the audio is the natural next source of evidence independent of the history.

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